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Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Half Past Paper – Week 11

Higher Tier

Total Marks

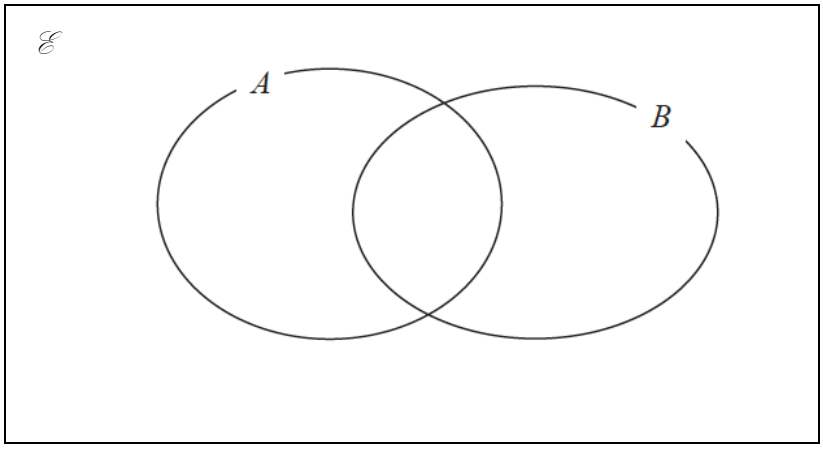
Q	Topic	Marks	Sparx Code
1a	Using set notation, Venn diagrams with set notation	3	U296,U748
1b	Using set notation, Venn diagrams with set notation	2	U296,U748
3a	Finding averages from frequency tables	1	U569
3b	Drawing and interpreting frequency polygons	2	U840
5	Angles in polygons	4	U427
7	Using a calculator, Finding unknown sides in right-angled triangles	2	U926,U283
9	Solving inverse proportion word problems	3	U357
11a	Multiplying and dividing numbers in standard form, Calculating with speed	2	U264,U151
11b	Multiplying and dividing numbers in standard form, Calculating with speed	1	U264,U151
13	Calculating with density	4	U910
15	Transforming graphs	2	U455
17	Identifying types of graphs	2	U593,U229, U450,U238
19a	Finding error intervals for calculations	3	U587
19b	Finding error intervals for calculations	2	U587
21	Interpreting histograms, Interpreting pie charts	4	U983,U172
23	Calculating with trigonometry and bearings, The sine rule, The cosine rule	5	U164,U952, U591
	Total	42	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
$A = \{1, 5, 6, 8, 9\}$
$B = \{2, 6, 9\}$



(*a*) Complete the Venn diagram to represent this information.

(3)

A number is chosen at random from the universal set $\mathscr{E}$.

(*b*) Find the probability that the number is in the set $A \cap B$

.....

(2)

(Total for Question 1 is 5 marks)

3 The table shows information about the heights of 80 plants.

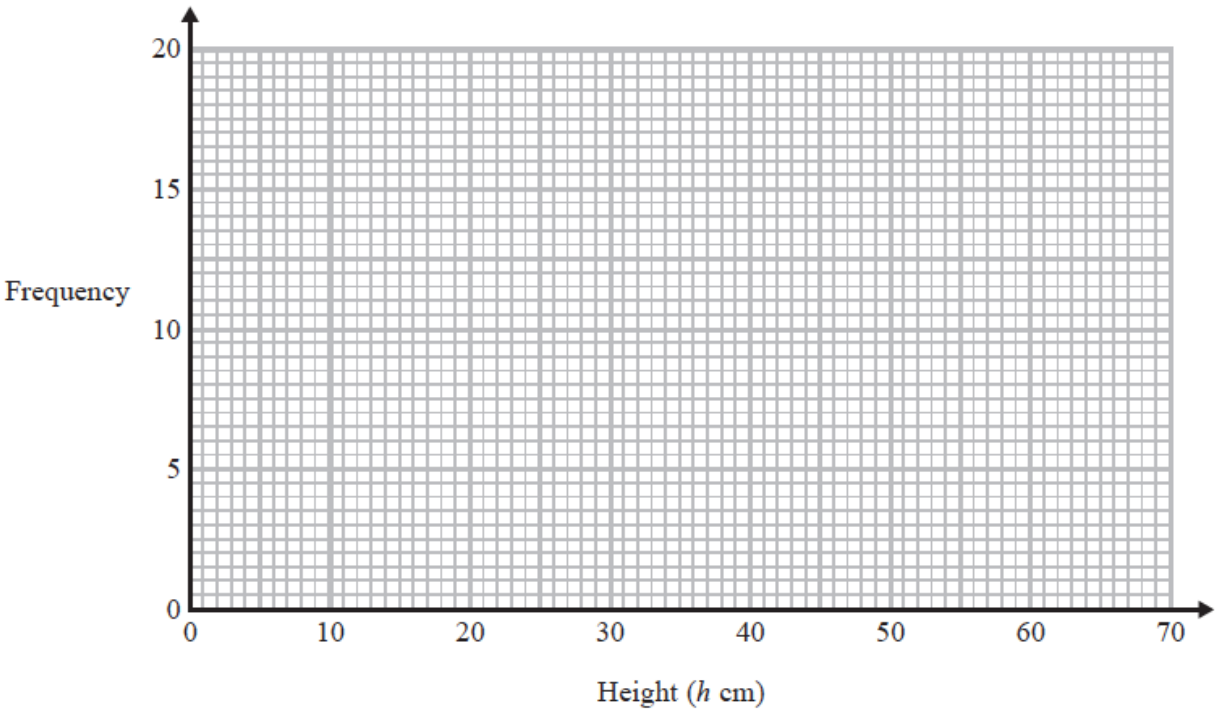
Height (h cm)	**Frequency**
$10 < h \leqslant 20$	7
$20 < h \leqslant 30$	13
$30 < h \leqslant 40$	14
$40 < h \leqslant 50$	12
$50 < h \leqslant 60$	16
$60 < h \leqslant 70$	18

(a) Find the class interval that contains the median.

.....

(1)

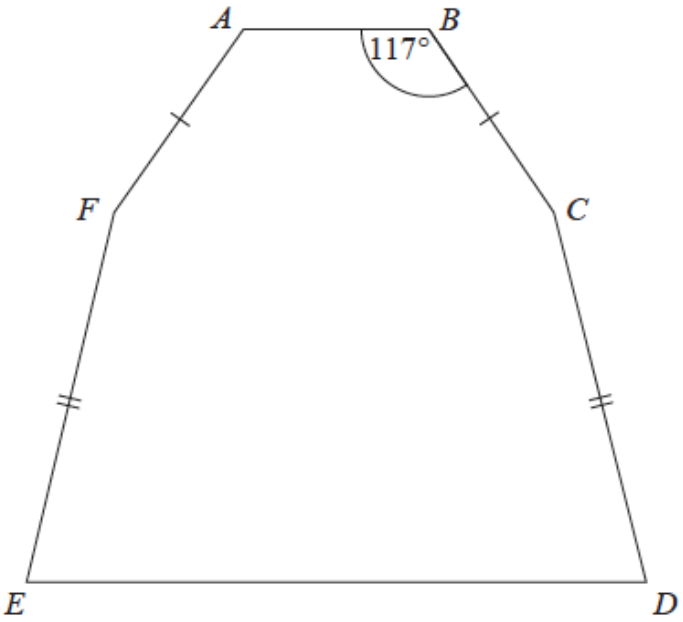
(b) On the grid, draw a frequency polygon for the information in the table.



(2)

(Total for Question 3 is 3 marks)

5 The diagram shows a hexagon.
The hexagon has one line of symmetry.



FA = *BC*
EF = *CD*
Angle *ABC* = 117°

Angle *BCD* = 2 × angle *CDE*

Work out the size of angle *AFE*.
You must show all your working.

.....°

(Total for Question 5 is 4 marks)

7 Work out $\sqrt{\dfrac{2.5 \times \sin 43^\circ}{8.2^2 - 50.5}}$

Give your answer correct to 3 significant figures.

.....

(Total for Question 7 is 2 marks)

9 A company has to make a large number of boxes.

The company has 6 machines.
All the machines work at the same rate.
When all the machines are working, they can make all the boxes in 9 days.

The table gives the number of machines working each day.

	day 1	day 2	day 3	all other days
Number of machines working	3	4	5	6

Work out the total number of days taken to make all the boxes.

.....

(Total for Question 9 is 3 marks)

11 In May 2019, the distance between Earth and Mars was 3.9×10^7 km.

In May 2019, a signal was sent from Earth to Mars.
Assuming that the signal sent from Earth to Mars travelled at a speed of 3×10^5 km per second,

(*a*) how long did the signal take to get to Mars?

..... seconds
(2)

The speed of the signal sent from Earth to Mars in May 2019 was actually less than 3×10^5 km per second.

(*b*) How will this affect your answer to part (*a*)?

.....

.....

.....
(1)

(Total for Question 11 is 3 marks)

13 The density of ethanol is 1.09 g/cm^3
The density of propylene is 0.97 g/cm^3

60 litres of ethanol are mixed with 128 litres of propylene to make 188 litres of antifreeze.

Work out the density of the antifreeze.
Give your answer correct to 2 decimal places.

..... g/cm^3

(Total for Question 13 is 4 marks)

15 The graph of the curve C with equation $y = f(x)$ is transformed to give the graph of the curve S with equation $y = f(-x) - 3$

The point on C with coordinates (7, 2) is mapped to the point Q on S.

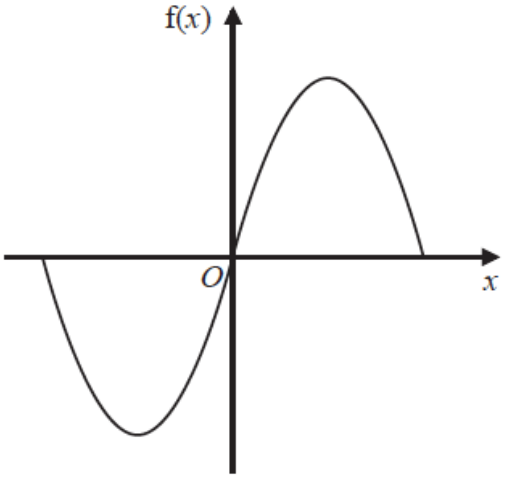
Find the coordinates of Q.

(..... , ...)

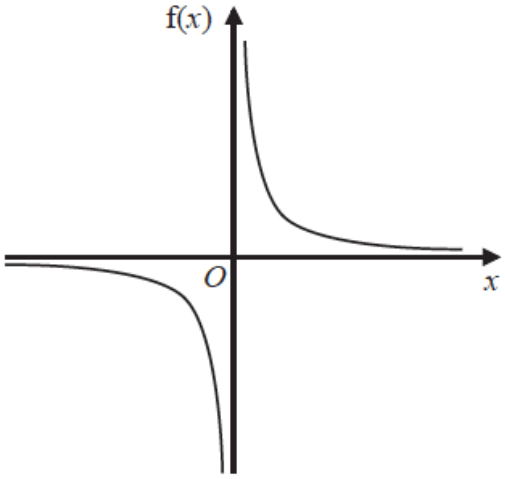
(Total for Question 15 is 2 marks)

17 Here are four graphs.

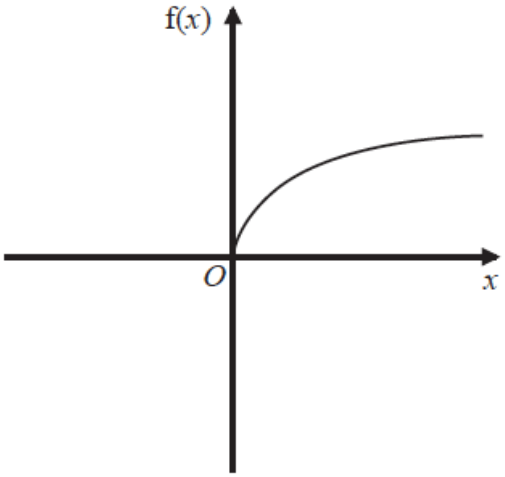
Graph A



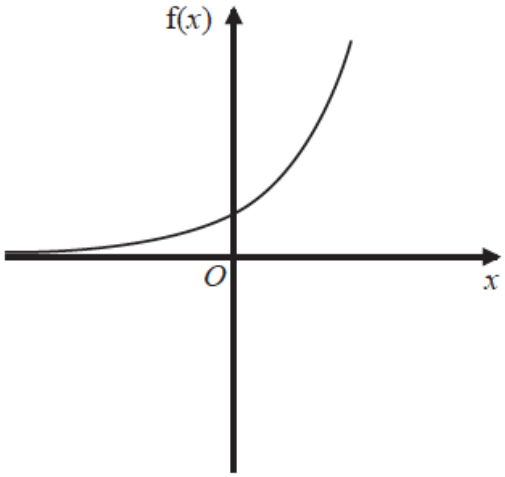
Graph B



Graph C



Graph D



The graphs represent four different types of function f.

Match each description of the function in the table to the letter of its graph.

Description of function	**Graph**
f(x) is inversely proportional to x	
f(x) is a trigonometrical function	
f(x) is an exponential function	
f(x) is directly proportional to $\sqrt{x}$	

(Total for Question 17 is 2 marks)

19 $D = \dfrac{u^2}{2a}$

u = 26.2 correct to 3 significant figures

a = 4.3 correct to 2 significant figures

(*a*) Calculate the upper bound for the value of D.
Give your answer correct to 6 significant figures.
You must show all your working.

.....

(3)

The lower bound for the value of D is 78.6003 correct to 6 significant figures.

(*b*) By considering bounds, write down the value of D to a suitable degree of accuracy.
You must give a reason for your answer.

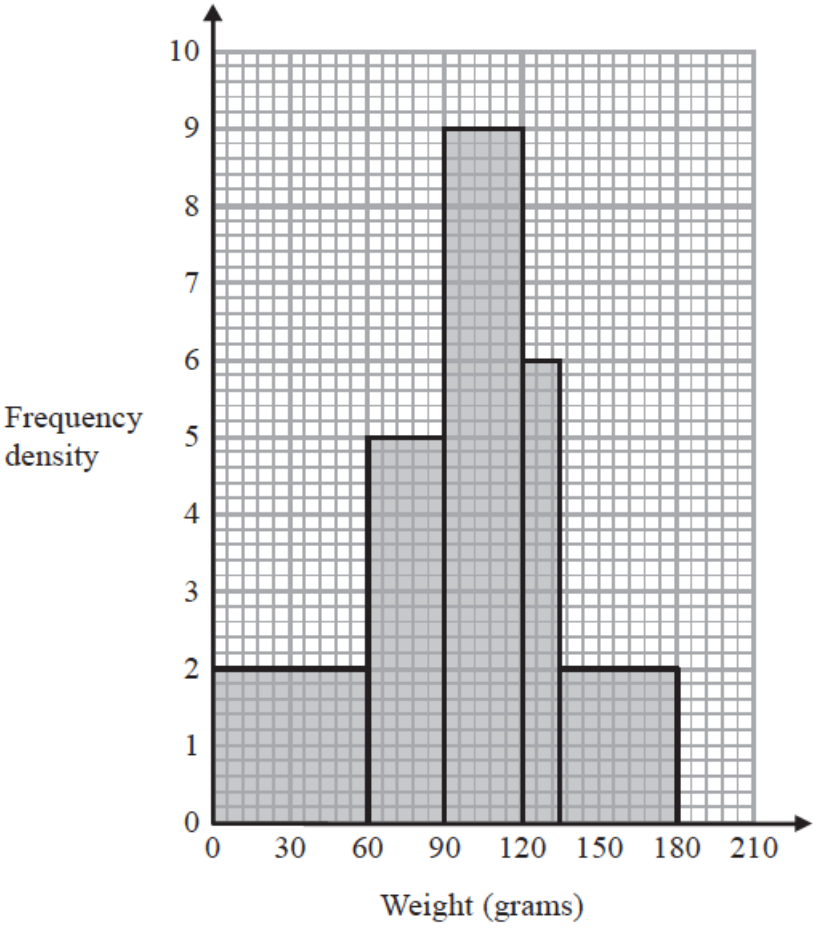
.....

.....

(2)

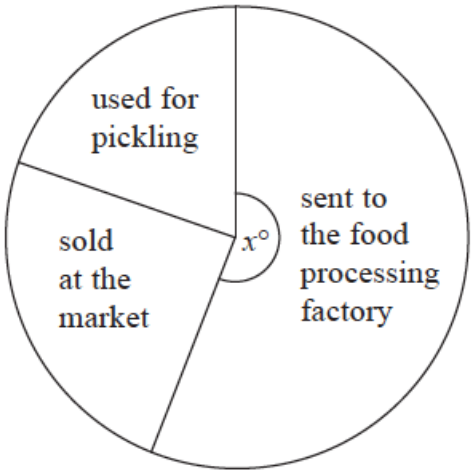
(Total for Question 19 is 5 marks)

21 The histogram gives information about the distribution of the weights of some onions grown by a farmer.



Onions less than 60 grams in weight are used for pickling.
Onions greater than 120 grams in weight are sold at the market.
The rest of the onions are sent to a food processing factory.

A pie chart is drawn using the information opposite to show what the farmer does with the onions he grows.



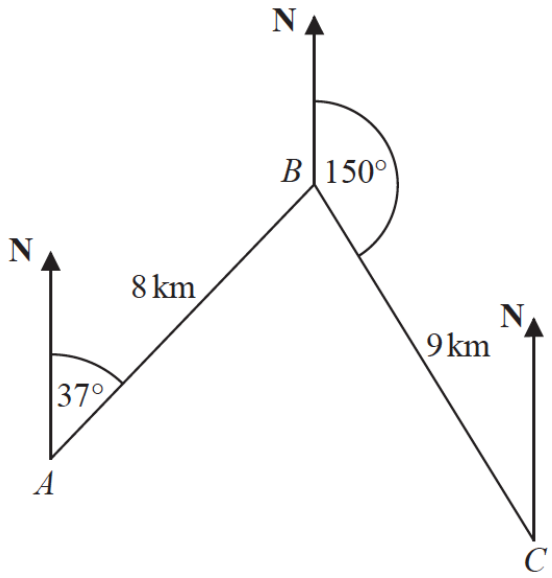
The angle of the sector for the onions sent to the food processing factory is $x°$.

Work out the value of x.

x = ...

(Total for Question 21 is 4 marks)

23 The diagram shows the positions of three towns, Acton (*A*), Barston (*B*) and Chorlton (*C*).



Barston is 8 km from Acton on a bearing of 037°
Chorlton is 9 km from Barston on a bearing of 150°

Find the bearing of Chorlton from Acton.
Give your answer correct to 1 decimal place.
You must show all your working.

.....°

(Total for Question 23 is 5 marks)

TOTAL FOR PAPER IS 42 MARKS